

Md. KAWSER TALUKDER

Machine Learning & Backend Engineer

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📍 Uttara, Dhaka-1230, Bangladesh

🌐 Portfolio

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🌐 LinkedIn

🐙 GitHub



PROFESSIONAL SUMMARY

Results-driven Machine Learning and Backend Engineer specializing in Deep Learning, Computer Vision, and highly scalable backend architectures. Demonstrated ability to process complex medical datasets, design high-accuracy explainable AI models, and engineer enterprise-grade full-stack platforms using Django REST Framework and ASP.NET Core MVC. Proficient in integrating secure payment gateways, implementing advanced machine learning pipelines, and resolving critical database concurrency bottlenecks. Passionate about bridging the gap between rigorous algorithmic research and reliable, data-driven software solutions for real-world applications.

TECHNICAL SKILLS

Programming

Python, C++, C, C#, Java, JavaScript, SQL.

Machine Learning & AI

Machine Learning, Deep Learning, Computer Vision, NLP, DenseNet121, Sentence-BERT, SVM, TensorFlow, Keras, PyTorch, Scikit-learn.

ML & Data Techniques

Data Preprocessing, Model Evaluation, Semantic Similarity, MixUp, Ordinal Encoding, Threshold Optimization, PCA, HOG, LBP, GLCM, SMOTE.

Data & Databases

NumPy, Pandas, Matplotlib, Seaborn, PostgreSQL, SQL Server, MySQL, Entity Framework Core.

Backend & Web

Django, Django REST Framework, ASP.NET Core MVC (.NET 8), React, Vite, REST APIs, JWT Authentication, Bootstrap, Axios.

Explainable AI & Tools

SHAP, LIME, Fairness Analysis, Git, GitHub, Docker, Jupyter Notebook, Google Colab.

Core CS

Data Structures & Algorithms, Object-Oriented Programming.

Language

Bengali: Native | English: Professional Working Proficiency | Hindi: Conversational.

PROFESSIONAL EXPERIENCE

Machine Learning Intern

Feb 2026 – Jun 2026

Career Point S@ifur's, Gazipur

- Executed end-to-end machine learning pipelines, focusing on Python-based data analysis, advanced preprocessing, and predictive model experimentation.
- Conducted rigorous feature engineering, model training, and performance evaluation on real-world datasets to solve practical classification challenges.
- Strengthened the ability to translate machine learning concepts and mathematical algorithms into deployment-ready software components.

KEY PROJECTS & RESEARCH

Diabetic Retinopathy Severity Classification

🐙 [GitHub Link](#)

Deep Learning, Computer Vision, TensorFlow, DenseNet121

- Engineered a computer vision classification pipeline processing 3,662 retinal fundus images to detect 5 progressive severity stages of Diabetic Retinopathy.
- Fine-tuned a DenseNet121 neural network architecture using transfer learning, achieving a peak **Validation Quadratic Weighted Kappa (QWK) of 0.9182**.
- Combated severe class imbalance (1,805 majority vs. 193 minority samples) by implementing **MixUp data augmentation** and an innovative **ordinal-style multi-label encoding**.
- Executed custom predictive threshold optimization (tuning from 0.5 to 0.4656) on the validation set, elevating the derived QWK to **0.9225** across the confusion matrix.
- Benchmarked the deep learning architecture against classical ML pipelines (Random Forest, XGBoost, CatBoost) utilizing handcrafted visual features (HOG, LBP, GLCM).

Doctor Appointment & Smart Healthcare Platform

[GitHub Link](#)

Full-Stack, React, Django, PostgreSQL, Gemini API

- Architected a highly scalable full-stack healthcare platform featuring complex Role-Based Access Control and multi-tier family-member management.
- Engineered an **AI-assisted symptom analysis engine** integrating the **Gemini API** with a custom rule-based fallback mechanism, enabling intelligent specialist recommendations based on symptoms, availability, and doctor ratings.
- Eliminated slot overbooking during high-traffic intervals by implementing PostgreSQL database transactions (`transaction.atomic()`) and row-level locking (`select_for_update()`).
- Integrated the **SSLCommerz** payment gateway via RESTful APIs for real-time transactions, alongside a policy-driven automated refund calculation algorithm.

Explainable Hybrid AI Framework for Resume Screening

[Colab Link](#)

NLP Research, Sentence-BERT, SHAP, LIME

- Designed an experimental AI framework capable of processing 2,482 real-world resumes across 24 distinct professional categories.
- Attained **77.95% classification accuracy** using a calibrated SVM classifier by transforming text into dense 384-dimensional semantic vectors via **Sentence-BERT**.
- Formulated a hybrid candidate ranking algorithm blending semantic similarity (60%), skill matching (20%), and model prediction confidence (20%) to optimize workflows.
- Incorporated **SHAP** and **LIME** for transparent decision-making (Explainable AI) and applied demographic parity metrics to evaluate and mitigate inherent bias.

Bus Ticket Booking & Transit Management System

[GitHub Link](#)

ASP.NET Core MVC, Entity Framework Core, C#

- Developed a scalable ASP.NET Core MVC web application for automated transit scheduling, dynamic seat selection, and session-based passenger management.
- Designed complex relational data models utilizing Entity Framework Core, establishing seamless foreign key relationships among passengers, buses, schedules, and bookings.
- Configured robust administrative workflows to streamline fleet management, dynamic routing, and live operational monitoring, secured by server-side C# Data Annotations.

CERTIFICATIONS, ACHIEVEMENTS & ACTIVITIES

- **Python Developer Certification (freeCodeCamp):** Completed approx. 300 hours of coursework and hands-on algorithmic Python development (Dec 2025). [\[View\]](#)
- **Machine Learning Specialization (Andrew Ng):** Comprehensive coursework covering Advanced Supervised and Unsupervised Learning algorithms. [\[View\]](#)
- **ML Bootcamp Certification:** Organized by IEEE CIS IUBAT SBC in collaboration with Shohoj Skill & Shohoj Coding (Aug 2026). [\[View\]](#)
- **Project Champion (IUBAT):** Won 1st place at the Robotics Club Project Showcase for developing a Python & Arduino-based Hand Finger Counter (Feb 2026). [\[View\]](#)
- **Dutch-Bangla Bank Scholarship Recipient:** Awarded for continuous academic excellence and outstanding merit. [\[View\]](#)
- **Grameen Bank Scholarship Recipient:** Recognized for outstanding academic performance. [\[View\]](#)
- **Contest Participant:** Competed in the IUBAT Junior Programming Contest (2023), significantly enhancing coding efficiency, algorithmic logic, and problem-solving skills.
- **Volunteer:** Supported event coordination and welcomed new students at the IUBAT Freshers Orientation Program (2023).

ALMA MATER

International University of Business Agriculture and Technology (IUBAT)

(May 2022- April 2026)

Bachelor of Science in Computer Science and Engineering (B.Sc. CSE) [↗](#)

CGPA: 3.50 / 4.00

Ibrahim Khan Govt. College, Bhuapur [\[View\]](#)

2020

Higher Secondary Certificate (H.S.C) [↗](#)

GPA: 5.00* / 5.00

Bhuapur Pilot High School, Bhuapur [\[View\]](#)

2017

Secondary School Certificate (S.S.C) [↗](#)

GPA: 5.00* / 5.00